

Writing Documentation with Free Software Tools

This article features popular free and open source tools for writing documentation for software.



Documentation is an integral part of creating an effective system, though it is sometimes neglected by many developers. This article discusses the Markup language for writing documentation, the tools used for writing the documentation, the conversion from one format to another, as well as hosting and versioning of documents. We will focus on Markdown and reST, and the tools for building the documents written using these. Many technical writers prefer LaTeX, since it is a high-quality typesetting system that includes features designed for the production of technical and scientific documentation. Markdown and reST are the programmers' choice because they come with many tools like Pandoc, Sphinx and ReadTheDocs, which assist in the conversion, building, versioning and hosting of the documents. LibreOffice is the choice of document writers who are usually not from a programming background, and is known for its ease of use.

Markdown

Markdown is a text format documentation language that can be used to easily document software and the systems developed with the key design goal of

readability. It was developed by John Gruber and Aaron Swartz. Modern editors like Vi, Atom, VScode and Web based versioning systems like GitLab and GitHub support Markdown. Typora is a Markdown editor with live preview functionality. The *README.md* file included in the source code of various free software is written in this lightweight Markdown format.

Markdown support for GNU Emacs can be enabled by using the following command in Debian based systems:

```
sudo apt-get install elpa-markdown-mode
```

Markdown syntax

Using # we can create headings, e.g., *#Welcome* is equivalent to `<h1>Welcome</h1>` in HTML. Likewise, we can create sub-headings, H2, H3, etc, using ## and ###. Alternatively, H1 and H2 can also be written as follows:

```
Heading1
=====
```

```
Heading2
-----
```

To create paragraphs, use blank lines to separate one or more lines of

text. Also, make sure that you do not indent paragraphs with spaces or tabs.

To create a line break (`<br>`), end a line with two or more spaces, and then type *return*.

Giving emphasis to text can be done using **text** or *_text_* for italics and ****text**** or **__text__** for making text bold. To get text in bold and italics, embed the text with trees or *_. eg __text__ or ***text***

Blockquotes can be created using the following syntax in Markdown:

```
> This is a blockquote.
>
> This is the second paragraph in the
blockquote.
>
> ## This is an H2 in a blockquote
```

Lists in Markdown can be generated in multiple ways -- using *, -, or numbered lists using #.

The syntax is as follows:

```
* This is a list
* This is an another list
* This is a sublist
* This is another sublist
```

can be replaced with - or + to create unordered lists and sub-lists. Simple ordered lists can be created with the following syntax (lines should start with the):

```
#. List item 1
#. List item 2
#. List item 3
```

Ordered lists with sub-lists can